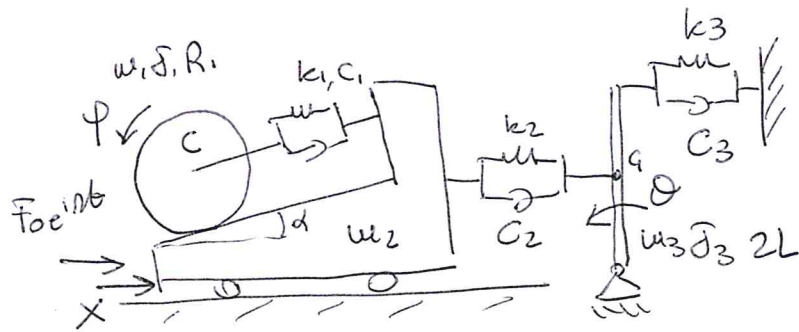


Simplified Test



$$x = \begin{Bmatrix} \phi \\ x \\ \theta \end{Bmatrix}$$

$$T = \frac{1}{2} m_1 \dot{x}_c^2 + \frac{1}{2} m_1 \dot{y}_c^2 + \frac{1}{2} J_1 \dot{\phi}^2 + \frac{1}{2} m_2 \dot{x}^2 + \frac{1}{2} m_3 \dot{x}_g^2 + \frac{1}{2} J_3 \dot{\theta}^2$$

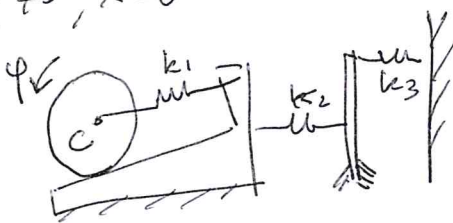
$$D = \frac{1}{2} \sum_{i=1}^3 c_i \Delta L_i^2 ; V_{el} = \frac{1}{2} \sum_{i=1}^3 k_i \Delta L_i^2 ; V_g = m_1 g h_{s1} + m_3 g h_{s3}$$

$$\delta W = F_{oeint} \delta x$$

$$[m_{PH}] = \text{diag}(m_1, m_1, J_1, m_2, m_3, J_3)$$

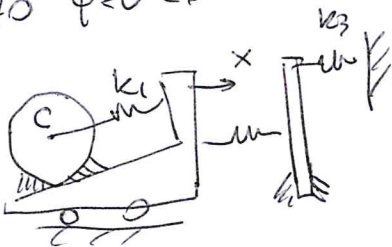
$$[c_{PH}] = \text{diag}(c_1, c_2, c_3) \quad [k_{PH}] = \text{diag}(k_1, k_2, k_3) \quad \underline{F} = \{ F_{oeint} \}$$

1) $\phi \neq 0 ; x = \theta = 0$



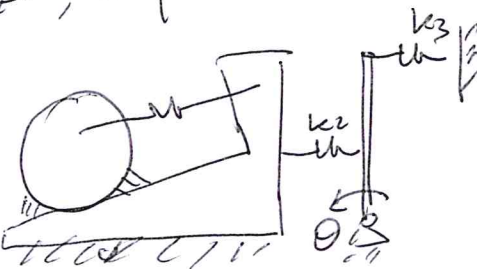
$$\begin{aligned} x_c &= -R\phi \cos \alpha & \Delta L_1 &= R\phi \\ y_c &= -R\phi \sin \alpha & \Delta L_2 &= 0 \\ x_g &= 0 & \Delta L_3 &= 0 \end{aligned}$$

2) $x \neq 0 \quad \phi = \theta = 0$



$$\begin{aligned} x_c &= x & \Delta L_1 &= 0 \\ y_c &= 0 & \Delta L_2 &= -x \\ x_g &= 0 & \Delta L_3 &= 0 \end{aligned}$$

3) $\theta \neq 0 ; x = \phi = 0$

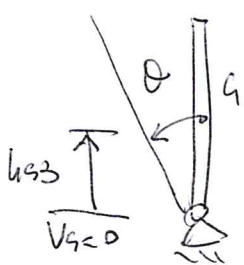


$$\begin{aligned} x_c &= 0 & \Delta L_1 &= 0 \\ y_c &= 0 & \Delta L_2 &= -L\theta \\ x_g &= -L\theta & \Delta L_3 &= 2L\theta \end{aligned}$$

$$\rightarrow \begin{cases} x_c = x - R\varphi \cos\alpha \\ y_c = -R\varphi \sin\alpha \\ x_g = -L\theta \\ \Delta l_1 = R\varphi \\ \Delta l_2 = -x - L\theta \\ \Delta l_3 = 2L\theta \end{cases} \quad [A_{ij}] = \begin{bmatrix} -R\cos\alpha & 1 & 0 \\ -R\sin\alpha & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -L \\ 0 & 0 & 1 \end{bmatrix}$$

$$[A_k] = [A_c] = \begin{bmatrix} R & 0 & 0 \\ 0 & -1 & -L \\ 0 & 0 & 2L \end{bmatrix} \quad [A_R] = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$$

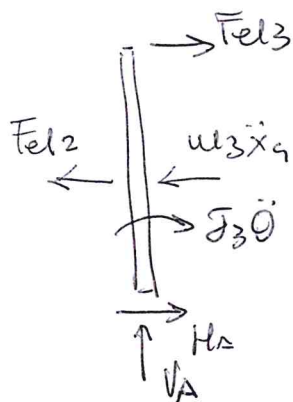
$$h_{q1} = y_c = -R\varphi \sin\alpha \rightarrow \text{depends linearly on } x \rightarrow [k_{q1}] = \neq$$



$$h_{q3} = L \cos\theta \rightarrow \frac{\partial^2 h_{q3}}{\partial \theta^2} = -L \cos\theta$$

$$[k_{q3}] = m_3 g \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -L \end{bmatrix}$$

• question 6



$$\sum F_x = 0 : H_A = F_{el2} - F_{el3} + m_3 \ddot{x}_q$$